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Single cell transcriptomic data using Rstudio

Data source:
https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE275330

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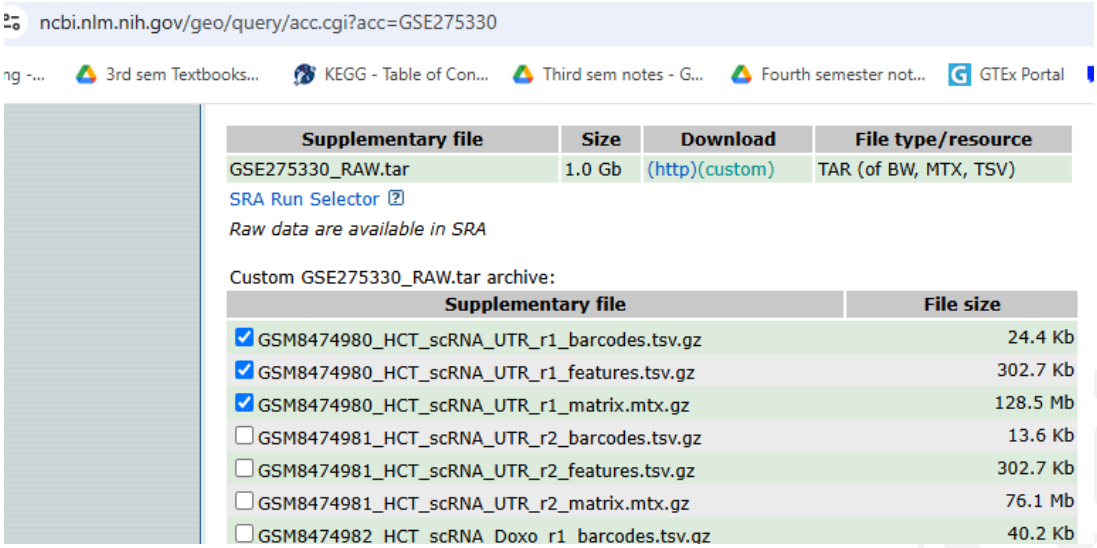
Scope: Self Format: HTML Amount: Quick GEO accession: GSE275330 GO

Series GSE275330 Query DataSets for GSE275330

Status	Public on Oct 09, 2024
Title	Doxorubicin resistance involves modulation of interferon signaling, transcriptional bursting, and gene co-expression patterns of U-ISGF3-related genes
Organism	Homo sapiens
Experiment type	Expression profiling by high throughput sequencing
Summary	we investigate the transcriptomic alterations induced by doxorubicin (DOX), a commonly used chemotherapeutic agent, in human colon cancer cells. Using single-cell RNA sequencing, we identified distinct cell populations and their transcriptional profiles following subtoxic dose DOX treatment, revealing cell clusters characterized by differential expression of genes involved in cell cycle regulation and interferon (IFN) signaling. We observed that expression of JAK/STAT pathway genes, activated by IFNs, did not correlate with expression of senescence markers. In contrast, DOX-persisting proliferating cells exhibited upregulation of genes induced by the unphosphorylated form of ISGF3 (U-ISGF3) transcription factor. Further analysis revealed that U-ISGF3 drives the expression of target genes by modulating the number of mRNA produced per transcriptional burst, potentially contributing to chemoresistance. Importantly, we found high upregulation of HSH2D, a poor prognostic marker, in doxorubicin-resistant U-ISGF3-dependent cells. Moreover, analysis of gene co-expression networks demonstrated that the altered network in DOX-treated cells, rather than individual gene expression levels, may dictate cancer cell fate after chemotherapy.




```
###SINGE CELL TRANSCRIPTOMICS ####

getwd()
setwd("G:/Rstudio/single_trans/")



#Load packages
library(Seurat)
library(dplyr)
library(Matrix)

#Install Seurat
install.packages(c("ggplot2", "dplyr", "Matrix", "cowplot", "Rcpp"))
install.packages("Seurat")
library(Seurat)

#Install Rtools 4.4
browseURL("https://cran.r-project.org/bin/windows/Rtools/")
system("g++ --version")

#Install dplyr
install.packages("dplyr", dependencies = TRUE)
library(dplyr)

#Install Matrix
```

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```
install.packages("Matrix")
library(Matrix)


#Install ggplot2
install.packages("ggplot2")
library(ggplot2)


str(seurat)     # Check the structure of your Seurat object
assay(seurat)   # List all assays in the Seurat object
names(seurat@assays$RNA@layers)   # Check layers available in the RNA assay
rna_counts <- seurat@assays$RNA@layers$counts  #access the count data
rna_data <- seurat@assays$RNA@layers$data   #access the normalized data



counts <- readMM("matrix.mtx.gz")
barcodes <- read.table("barcodes.tsv.gz", stringsAsFactors=F)[,1]
features <- read.csv("features.tsv.gz", stringsAsFactors=F, sep="\t", header=F)
rownames(counts) <- make.unique(features[,2])
colnames(counts) <- barcodes
counts


seurat <- CreateSeuratObject(counts, project="hb1")
#Warning: Feature names cannot have underscores ('_'), replacing with dashes ('-')
#Warning: Data is of class dgTMatrix. Coercing to dgCMatrix
#Just ignore this error
# Check the Seurat object summary
seurat



seurat[["percent.mt"]] <- PercentageFeatureSet(seurat, pattern = "^MT[-\\.]")
VlnPlot(seurat, features = c("nFeature_RNA", "nCount_RNA", "percent.mt"), ncol = 3)
```

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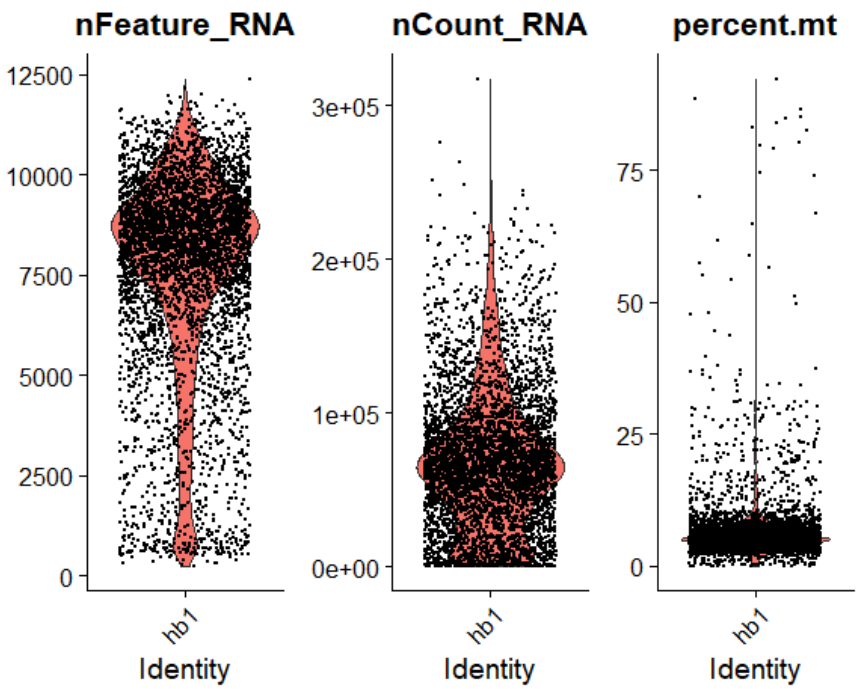
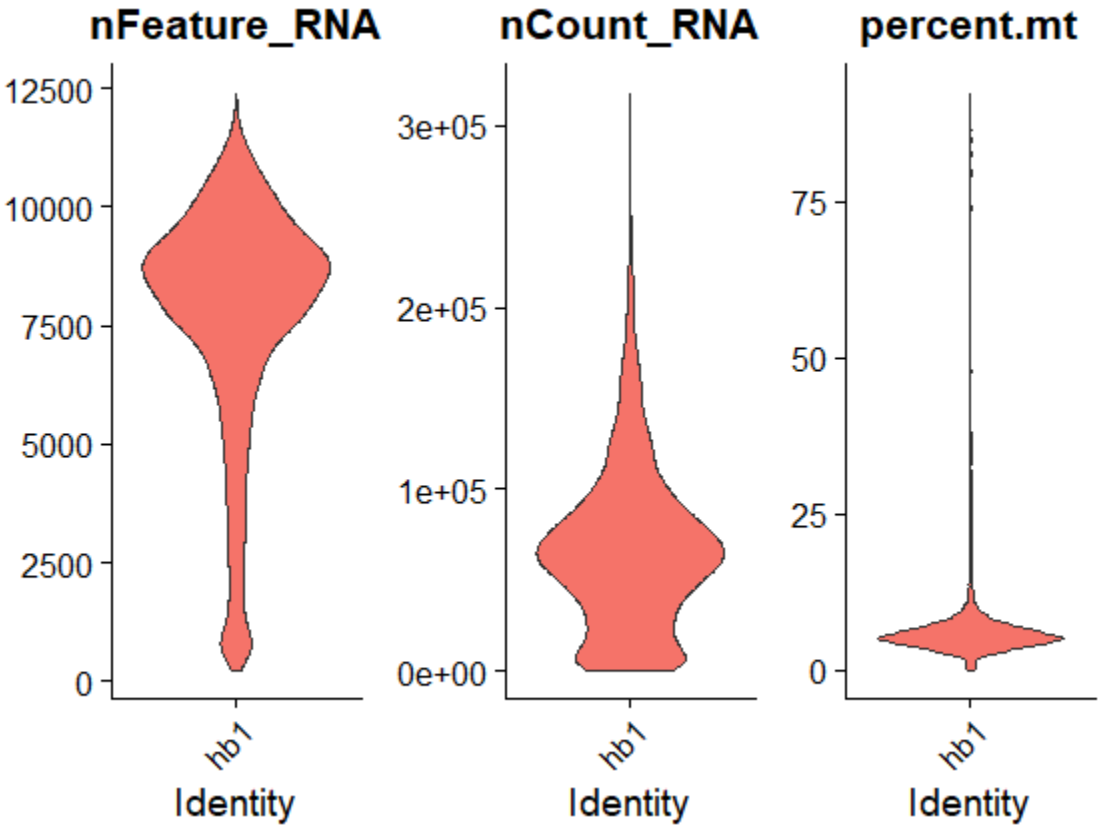
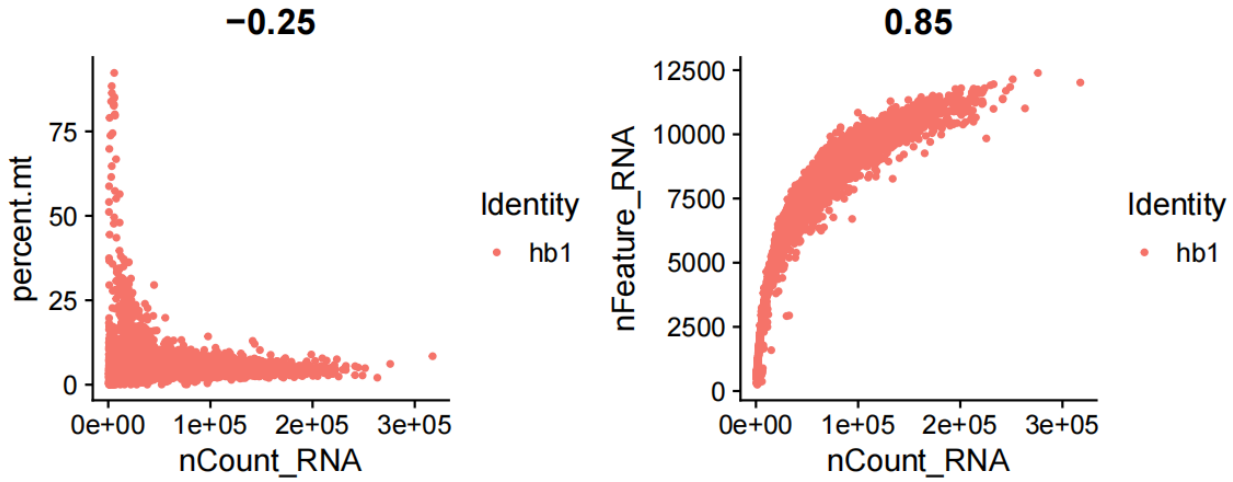


VlnPlot(seurat, features = c("nFeature_RNA", "nCount_RNA", "percent.mt"), ncol = 3, pt.size=0)



```
library(patchwork)
plot1 <- FeatureScatter(seurat, feature1 = "nCount_RNA", feature2 = "percent.mt")
plot2 <- FeatureScatter(seurat, feature1 = "nCount_RNA", feature2 = "nFeature_RNA")
plot1 + plot2
```




```
seurat <- subset(seurat, subset = nFeature_RNA > 500 & nFeature_RNA < 5000 & percent.mt <
5)
seurat <- NormalizeData(seurat)
seurat <- FindVariableFeatures(seurat, nfeatures = 3000)

top_features <- head(VariableFeatures(seurat), 20)
plot1 <- VariableFeaturePlot(seurat)
print(plot1)

plot2 <- LabelPoints(plot = plot1, points = top_features, repel = TRUE, xnudge = 0, ynudge = 0,
max.overlaps = 40, width = 10, height = 8, dpi = 300)
print(plot2)
```

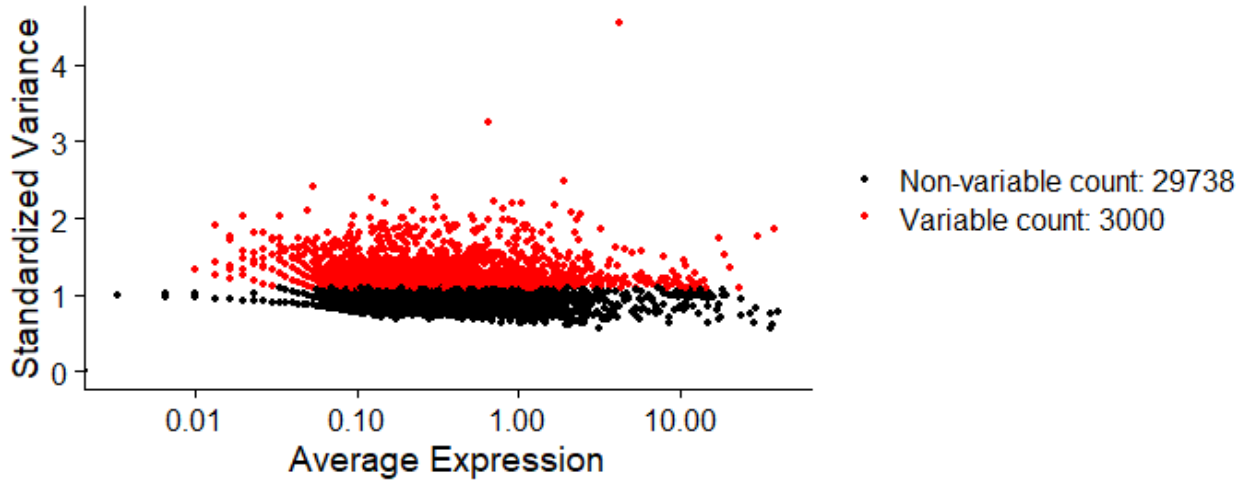


Fig: plot1

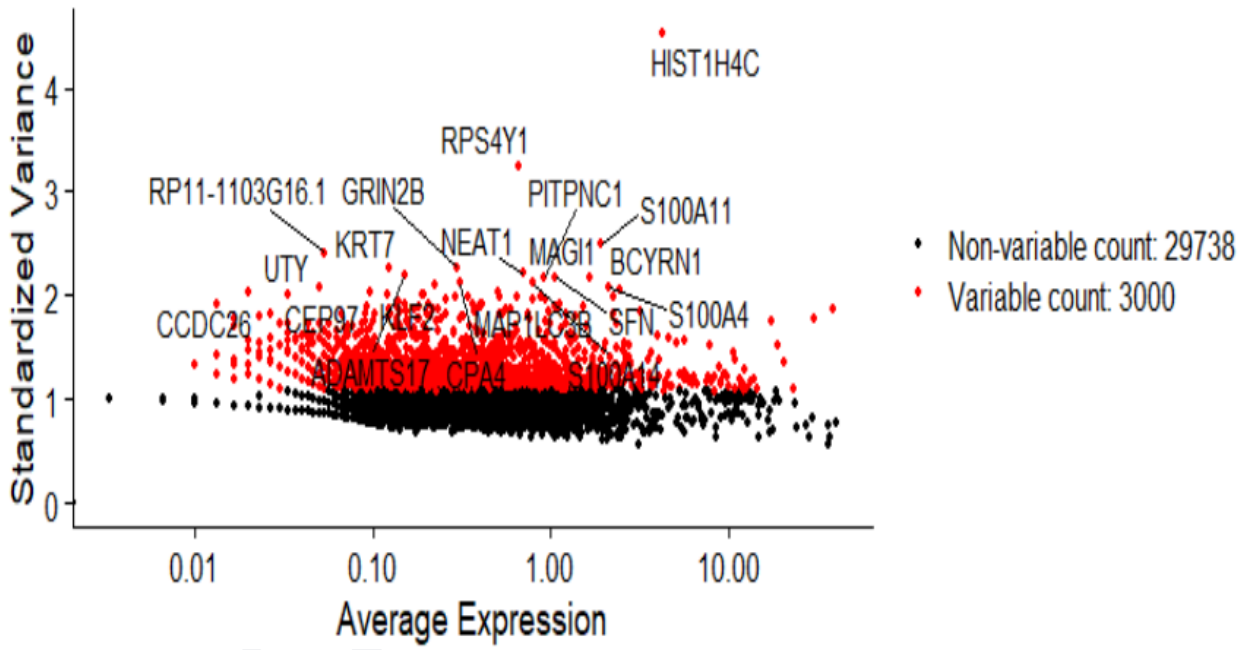


Fig: Plot2

Data scaling
seurat <- ScaleData(seurat)

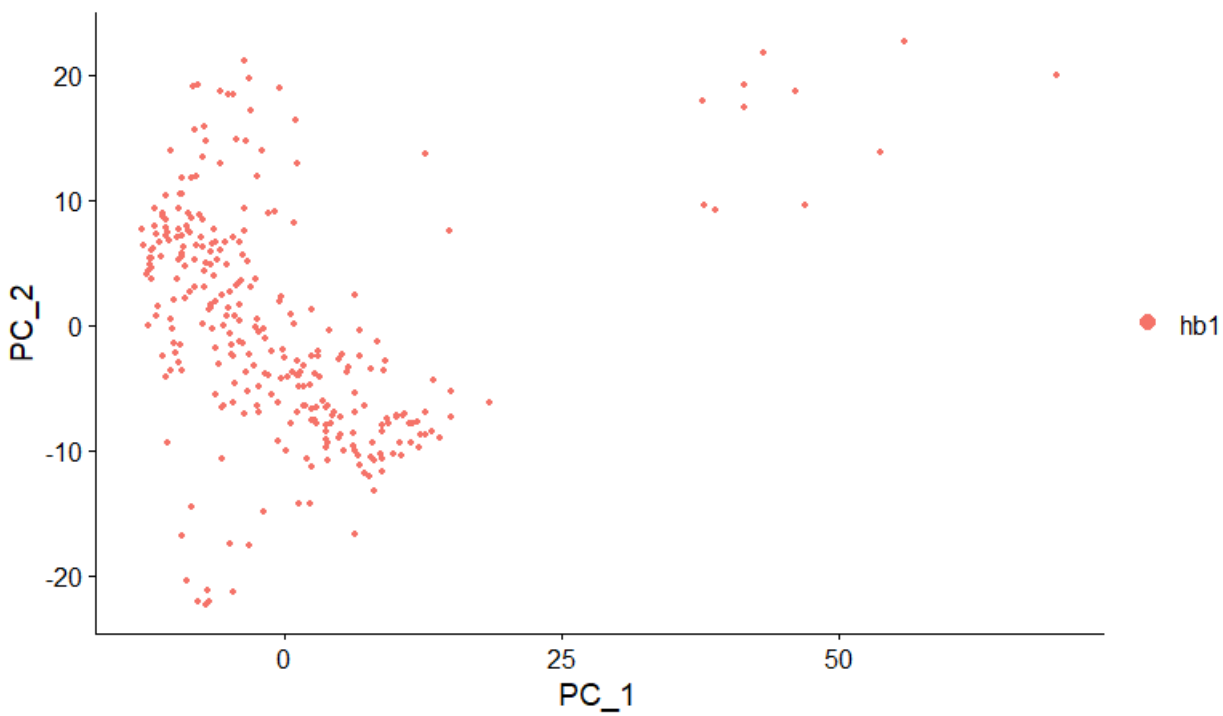
Centering and scaling data matrix
seurat <- RunPCA(seurat, npcs = 50)

@@Output—---
PC_ 1

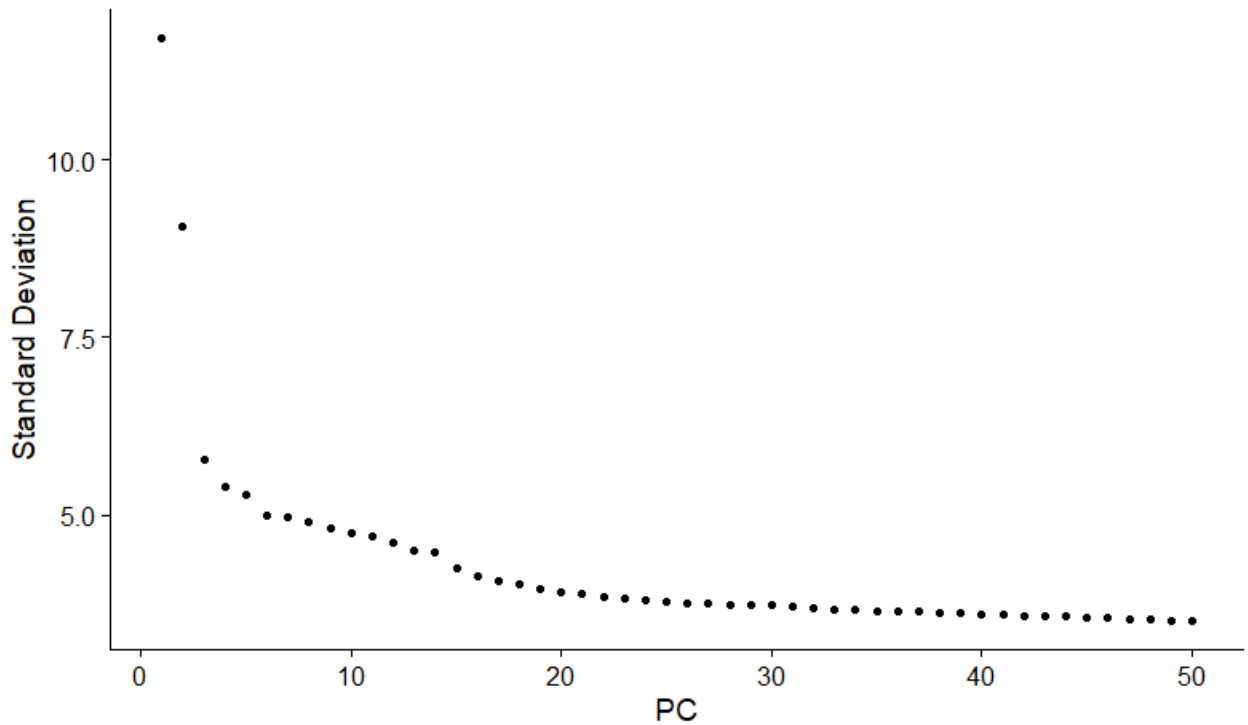
Positive: AC004158.2, DST, AGAP1, IGF2BP3, PITPNC1, VPS13B, ROR2, SLC35F3, PARD3, MBNL1
EHMT1, DNAH14, MICAL3, AGO2, TRAPPC9, UPK3B, CDC42BPA, STOX2, KDM3A, KMT2C
FRMD5, ARSG, HACE1, TBCD, ATP9A, FAF1, DOCK5, C11orf30, SETD5, C10orf76
Negative: ACTG1, S100A6, PRDX1, RPS27A, FTH1, TMSB10, GAPDH, RPL28, GNB2L1, FAU
RPS3A, RPS23, RPL32, NACA, FTL, TUBA1B, HMGA1, RPL10, RPL36, RPL15
RPS4X, RPL35A, UBB, EIF4A1, H3F3B, H2AFZ, MT2A, KRT18, RPL26, KRT19
PC_ 2
Positive: PLEC, MALAT1, CEP97, CHTF18, TAF1C, RHBDF2, BOP1, UTY, ARHGEF10, PKD1
FBLN5, ANKRD36C, KCNH3, RP11-115J23.1, KLHL17, SEMA3A, AP5Z1, RP11-264C15.2, PLEKHG4, PITPNM2
CCDC26, ARL17A, BTN2A1, L3MBTL3, RBM4, TMC6, SMARCAL1, GRIK4, BACH2, AVIL
Negative: FTL, RPL29, RPL35A, SLIRP, TMSB10, RPS23, FTH1, MT2A, H2AFZ, RPS15
MT1E, HSPE1, RPL32, PFN1, PRELID1, RPS10, H3F3B, RPS8, RPS5, RPS13
TPT1, RPL15, RPL10, UBA52, SPCS1, FABP5, FAU, PPP1R14B, RPL26, UQCR10
PC_ 3
Positive: BTD, WDR64, RP11-843A23.1, KCNV2, TTN, AVIL, NRP1, RP11-69E11.8, MARCH1, DLEU7
AC009313.1, RP11-220C2.1, STARD8, RP11-557C18.3, GRIA3, C10orf129, ECM2, TGFB2, LINC00959, RP11-739N20.2
TMLHE-AS1, Z83001.1, FMNL3, DIAPH3-AS1, RP11-554A11.8, FAM101A, NRG1, RP11-58G13.1, RP11-463D19.1, AADAC
Negative: FBLN5, MTMR11, CTD-2540F13.2, ARID4A, RNF165, WDR62, NFATC1, PRKD2, CTD-2537I9.12, EP400NL
KIAA0141, RECQL4, SNED1, MACROD1, DOK4, VCAN-AS1, NBEAL2, DEPDC5, RP11-387D10.2, JAK2
MLKL, PLCG1, NBPF9, GYS1, TRIB2, PABPC1L, RP13-188A5.1, CLIP2, LINC00640, DAGLB
PC_ 4
Positive: DOK4, C20orf112, JAZF1, HSH2D, AKNAD1, CCPG1, FAM227B, LRP4, SNED1, ZNF2
BACH2, ARID4A, C12orf79, MTMR11, RBM44, CATSPERD, IL15, CTD-2540F13.2, ST3GAL2, TP73
CNTROB, TRIM2, AVIL, SLC38A8, GORASP1, SELO, CLIP2, RP11-371F15.3, CTD-2537I9.12, FBXL20
Negative: CCDC26, SEMA3A, UTY, RP11-206L10.3, RP3-323P13.2, BCL2, CTD-2555O16.1, TTTY14, CLIP3, PRDM16
CGREF1, WASF3, RP11-269F21.3, NPIPB5, SLC27A6, FN3K, SUSD1, BBS9, HLF, NEURL1B
TARS2, AC092594.1, RAVER1, PDPR, AMH, AL162759.1, RBPMS2, GPR39, C1orf61, ADAMTS17
PC_ 5
Positive: RP5-1044H5.1, GORASP1, C12orf79, AC091153.4, CTD-2201G3.1, IL15, RP11-297P16.3, MCOLN3, TRAF3IP2-AS1, CATSPERD

```
        RBM44, NPAS1, ATXN1, RP11-4F22.2, AP000487.6, RNF43, RP11-355F16.1,
PPOX, ITPKB, ZNF383
        RP1-1J6.2, BCAS3, AC007098.1, RP11-118M9.3, MALAT1, RP11-472M19.2,
CD109, RASSF8, RC3H1, KIAA0319
Negative:  LRP4, FGD1, ATP8B2, MRAS, CCPG1, KIAA1462, FAM227A, GRIK4, FBLN5,
GMDS-AS1
        RP11-264C15.2, NBEAL2, ST3GAL2, CAMKK2, NBPF9, RNF165, EP400NL,
MTMR11, NINL, SLC38A8
        LDLRAD4, CTD-2540F13.2, ANKS1B, TUBG2, RP11-371F15.3, RPS6KA2, TRIM35,
CTD-2537I9.12, MED25, PGBD1
```

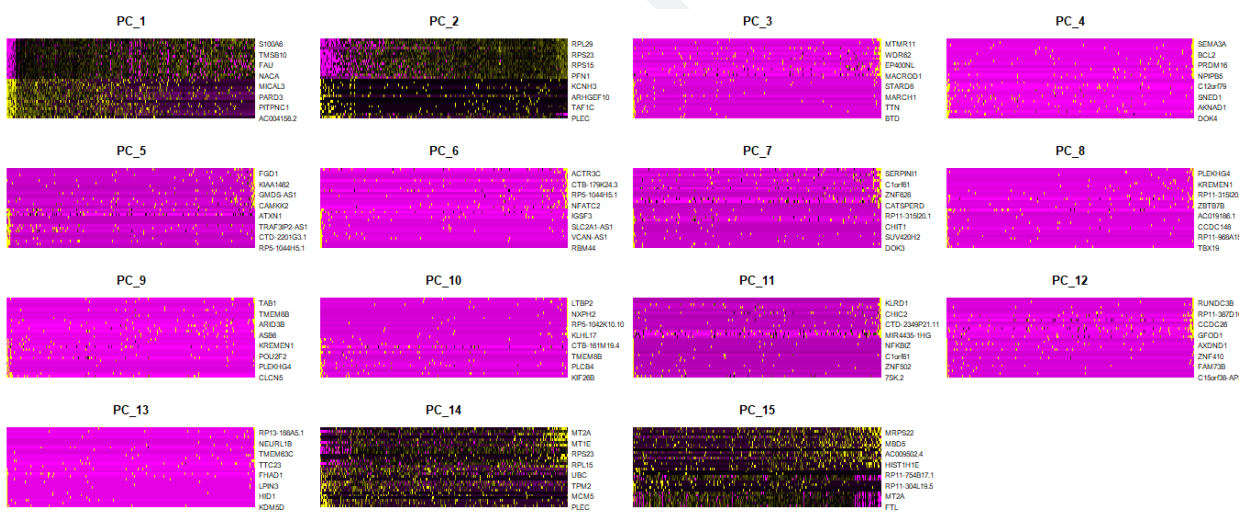
# Plotting the PCA visualization
pca_plot <- DimPlot(seurat, reduction = "pca", dims = c(1, 2)) # First and second PCA components
print(pca_plot)



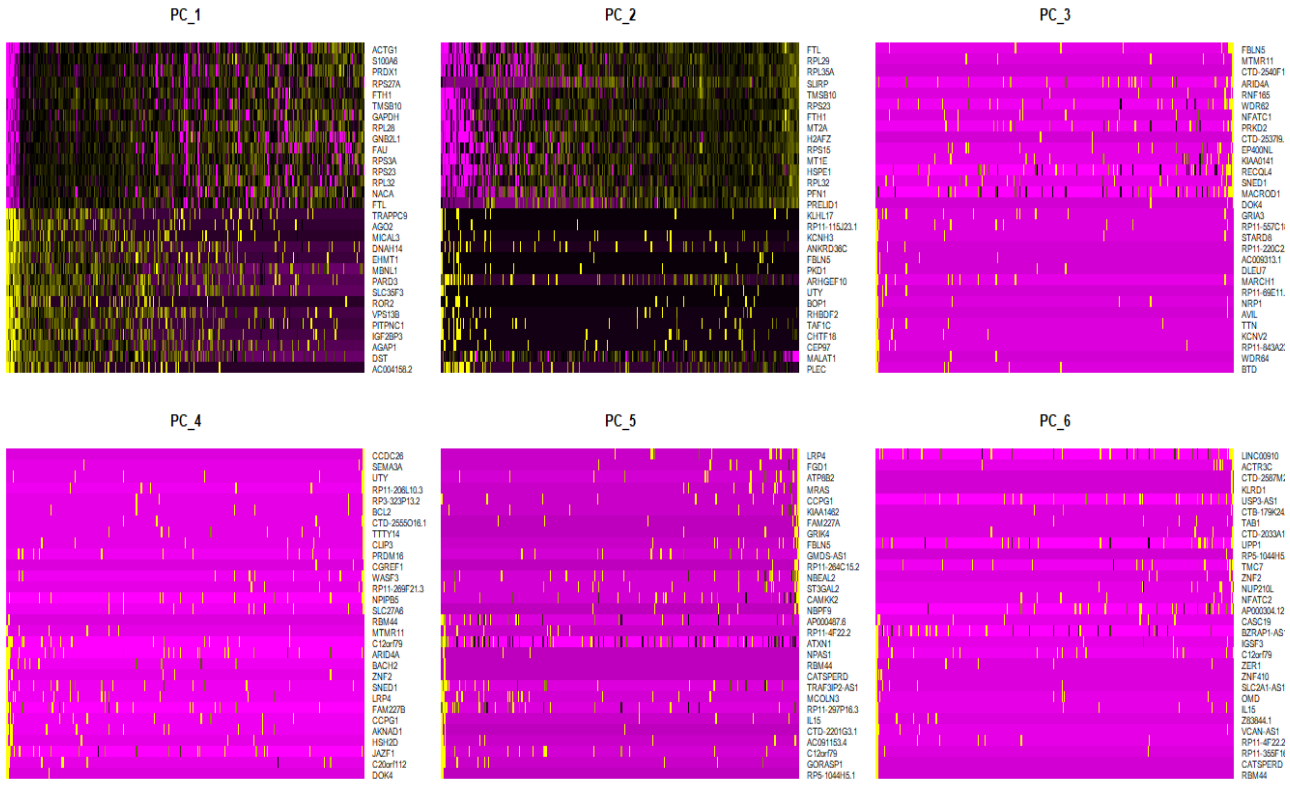
Elbow plot to choose number of PCs
ElbowPlot(seurat, ndims = ncol(Embeddings(seurat, "pca")))



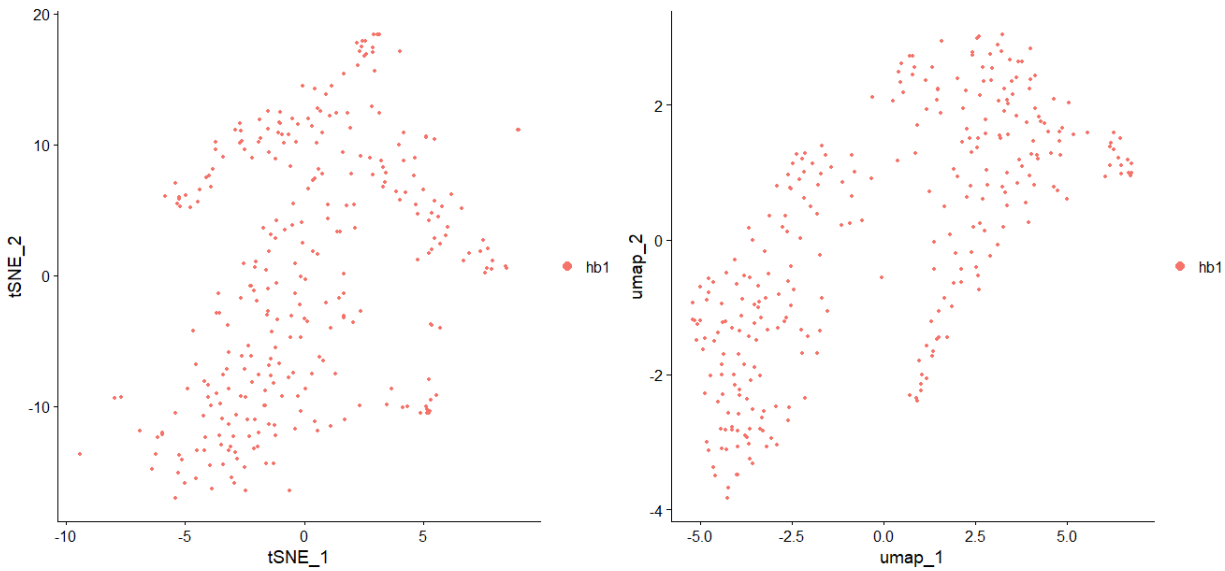
PCHeatmap(seurat, dims = 1:15, cells = 500, balanced = TRUE, ncol = 4)



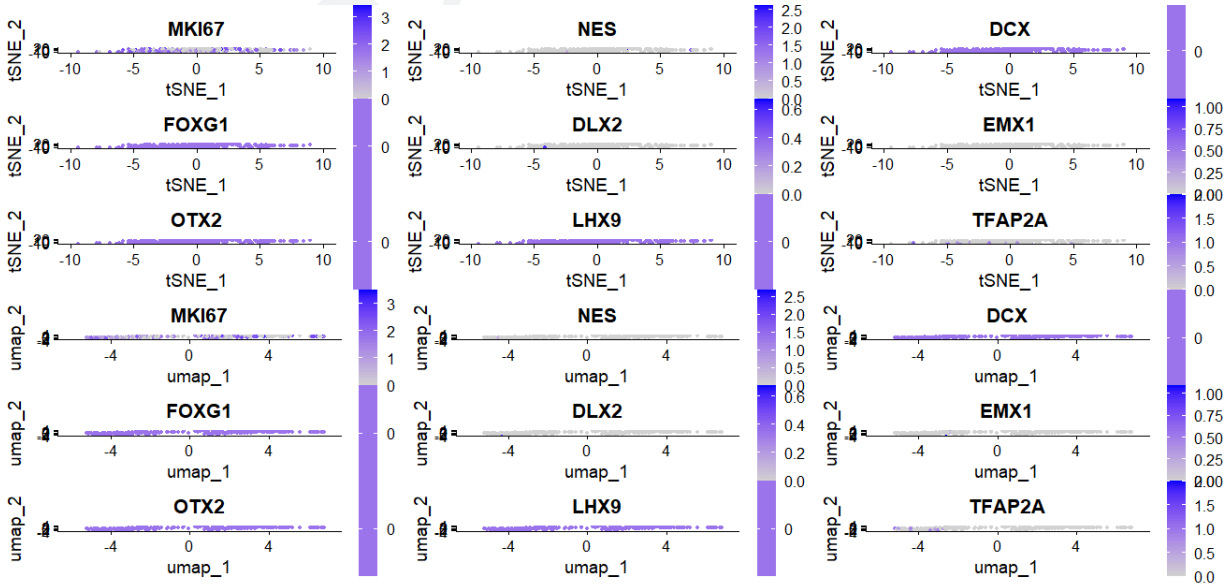
# Generate PC heatmaps and arrange them in 3 columns, 2 rows
pc_heatmap <- PCHeatmap(seurat, dims = 1:6, cells = 500, balanced = TRUE) # Showing 6 PCs
pc_heatmap + plot_layout(ncol = 3, nrow = 2)



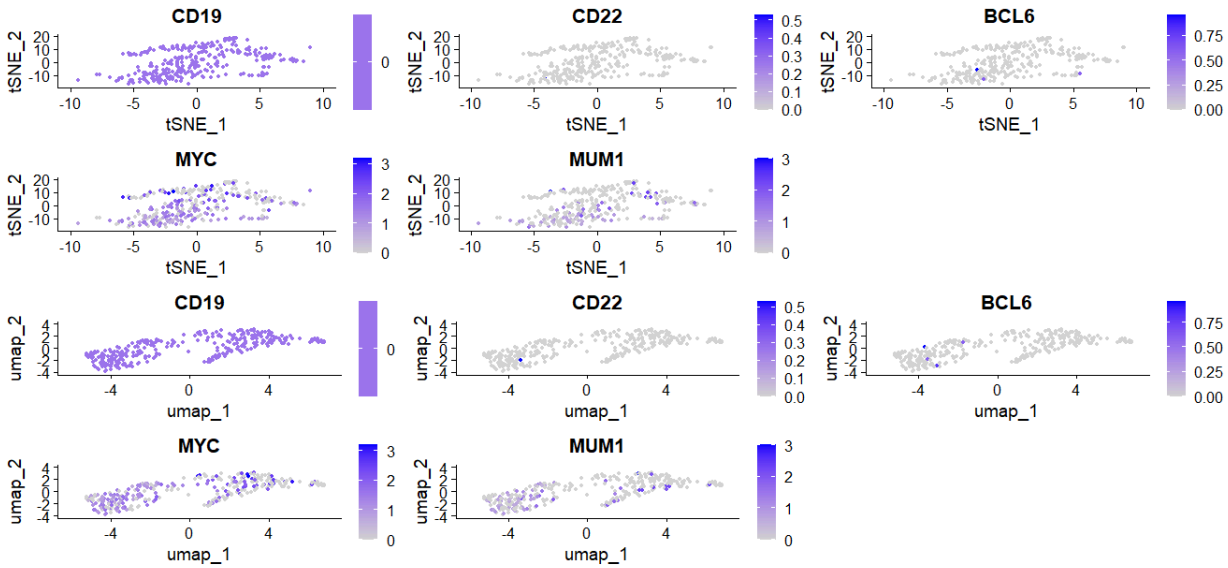

```
#### Clustering based on selected PCs
seurat <- RunTSNE(seurat, dims = 1:15)
seurat <- RunUMAP(seurat, dims = 1:15)

## plots
plot1 <- TSNEPlot(seurat)
plot2 <- UMAPPlot(seurat)
plot1 + plot2
```



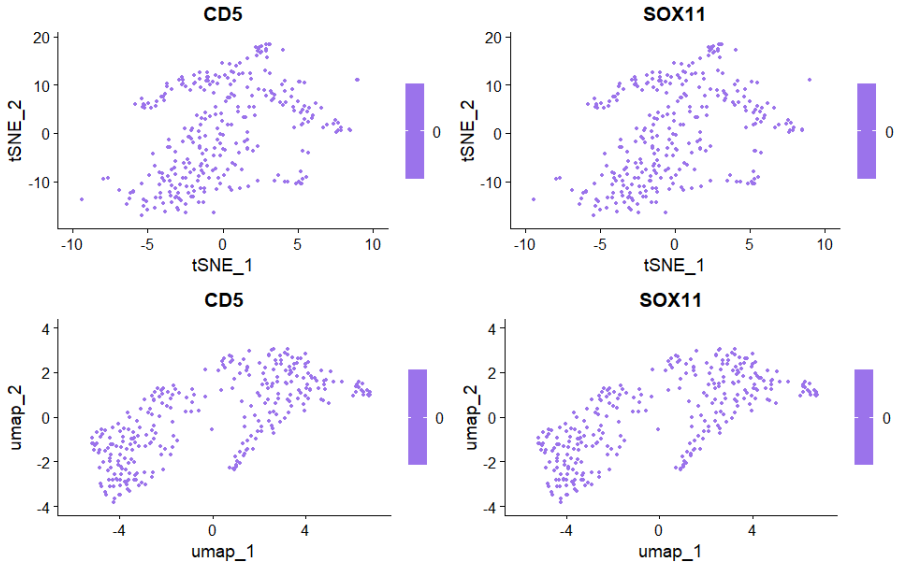

```
## checking and plotting canonical markers of the cell types
plot1 <- FeaturePlot(seurat,
c("MKI67","NES","DCX","FOXG1","DLX2","EMX1","OTX2","LHX9","TFAP2A"),
                ncol=3, reduction = "tsne")
plot2 <- FeaturePlot(seurat,
c("MKI67","NES","DCX","FOXG1","DLX2","EMX1","OTX2","LHX9","TFAP2A"),
                ncol=3, reduction = "umap")
plot1 / plot2
```



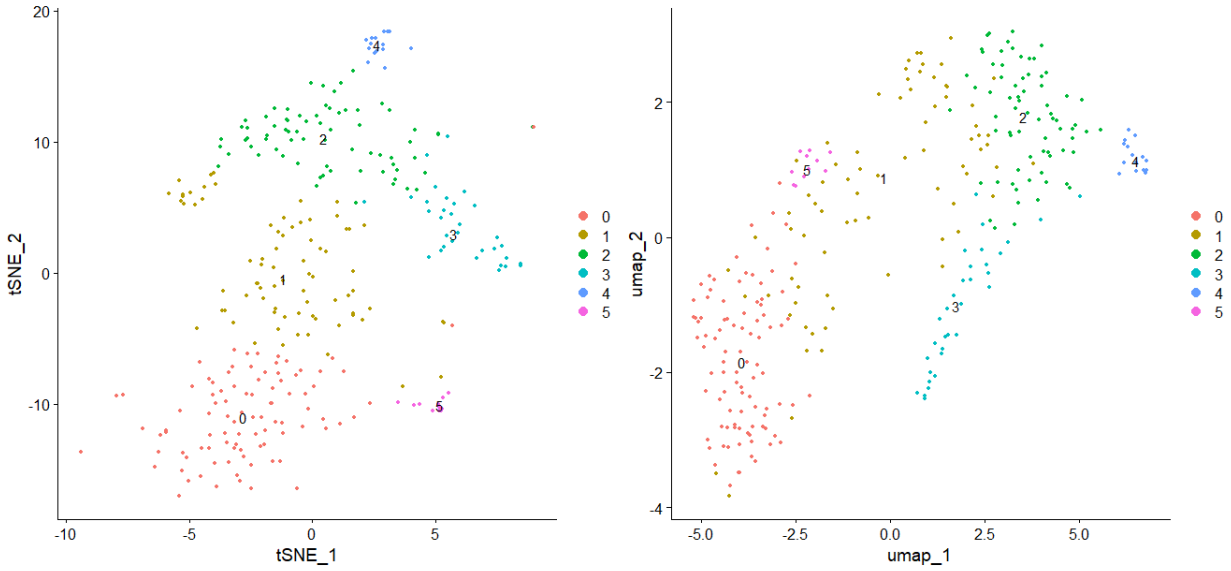
```
#Diffuse Large B-Cell Lymphoma (DLBCL) markers
plot1 <- FeaturePlot(seurat, c("CD19", "CD20", "CD22", "CD79a", "BCL6", "MYC",
"MUM1"),
                ncol=3, reduction = "tsne")
plot2 <- FeaturePlot(seurat, c("CD19", "CD20", "CD22", "CD79a", "BCL6", "MYC",
"MUM1"),
                ncol=3, reduction = "umap")
plot1 / plot2
```




```
#Mantle Cell Lymphoma (MCL)
plot1 <- FeaturePlot(seurat, c("CD5", "Cyclin D1", "SOX11"),
                ncol=3, reduction = "tsne")
plot2 <- FeaturePlot(seurat, c("CD5", "Cyclin D1", "SOX11"),
                ncol=3, reduction = "umap")
plot1 / plot2
```

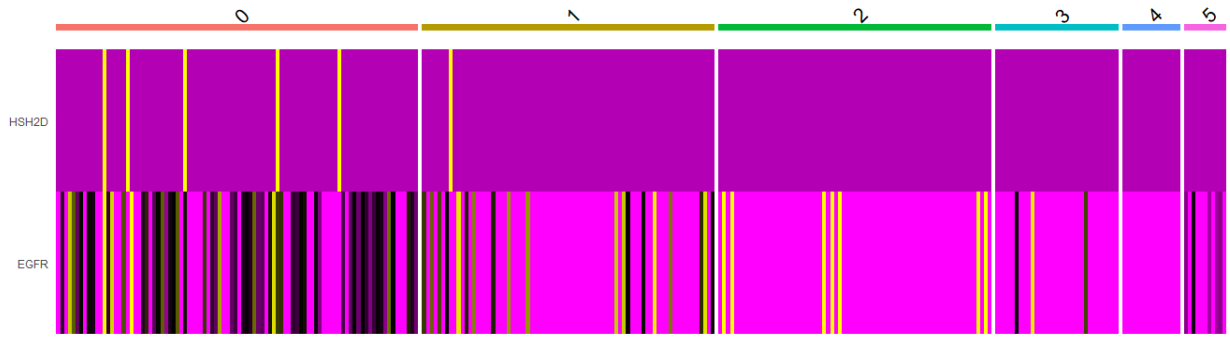


tSNE & UMAP results
plot1 <- DimPlot(seurat, reduction = "tsne", label = TRUE)
plot2 <- DimPlot(seurat, reduction = "umap", label = TRUE)
plot1 + plot2



# Prior knowledge of markers (from database/ Publications). These markers here are just used as an example.
colon_markers <- c("CEA", "CA 19-9", "mSEPT9", "FIT", "ctDNA", "KRAS", "BRAF", "TP53", "MSI-H", "HSH2D", "CDX2", "EGFR", "NRAS", "dMMR", "HER2", "CMS classification")

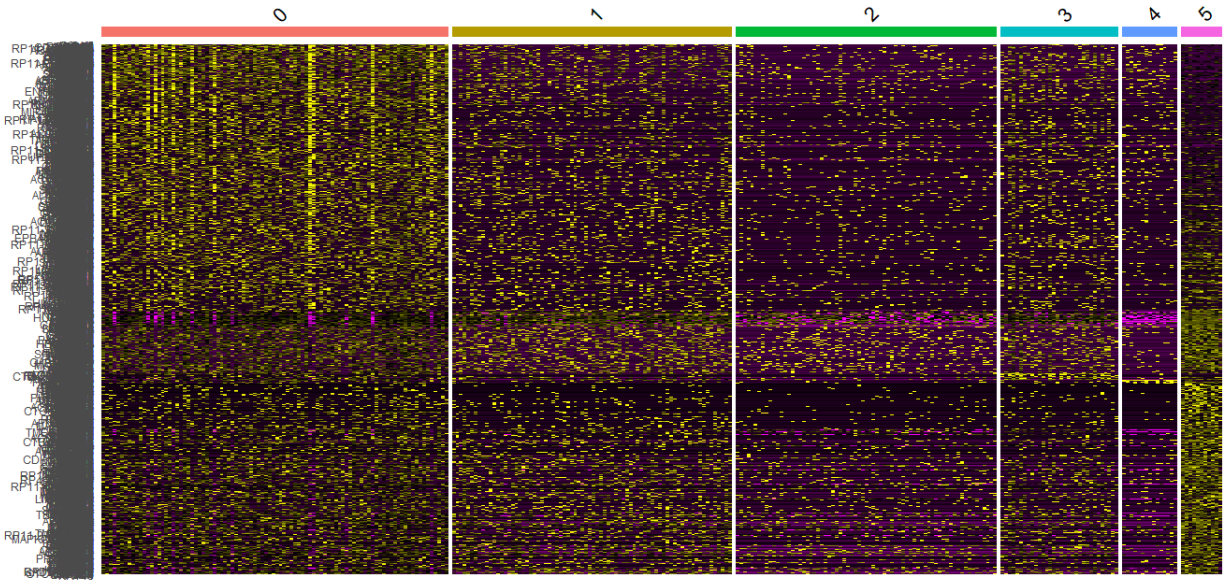
DoHeatmap(seurat, features = ct_markers) + NoLegend()



# Heatmap for visualizing identified markers
DoHeatmap(seurat, features = cl_markers$gene) + NoLegend()

#@output-

```
Warning message:
In DoHeatmap(seurat, features = cl_markers$gene) :
  The following features were omitted as they were not found in the scale.data
slot for the RNA assay: KLK6, EMP1, EBNA1BP2, TMEM179B, NECAP1, ING2, C19orf10,
ELAC1, GINS3, URGCP, ARFGAP3, ZBTB5, GTPBP1, STX7, NUDT5, C21orf33, CSTF2T,
TBL3, FAM207A, TIA1, METTL23, ZNF317, TICAM1, WDR45, PDCD5, C20orf27, CEBPD,
KCTD2, GRINA, RPAIN, EIF2S2, LSM7, NDUFAF4, VAC14, MIS18A, AASDHPPT, TLN1,
PITHD1, KLF10, RBP1, SUMO1, SRP9, C6orf120, ANKRD9, FASTKD5, ADAP2, STX4,
SEC24D, AMPD2, C16orf62, PROSER1, ZNF7, TMX2, ANKZF1, AC072062.1, CLPTM1,
PROCR, MAP3K11, VPS51, STK17A, TXNL4B, AP4M1, TRMT112, PHLDA1, NADSYN1,
C1GALT1C1, TPRA1, RSL24D1, CARD10, AAMDC, PSMB7, SLC25A4, UBE2T, PITRM1, LETM2,
MRPS18C, CDC6, RABL3, HSDL1, AMOTL2, CDCA4, PPAP2C, GPATCH11, PBDC1, EPHX1,
SLC25A1, ACADM, C6orf136, RCOR3, MFAP2, VPS29, CCDC88C, FAM127A, RPL39,
HCFC1R1, METTL6, DCAF17, PMF1, ITGAE, PSMC6, EHD1, RP6-109B7.3, AAED1, PES1,
ACOT7, C21orf91, RTN4R, KLC3, NEK3, MFSD5, RIC8A, AIF1L, SLC16A14, KTI12, PAX8,
CSRP1, TME [... truncated]
```



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